

COMPUTATIONAL HEAT AND FLUID FLOW

ASSIGNMENT - 2

1

Name: JAYATIRTHA HARIVITTAL MANGALVEDEKAR

Roll-No.: ME245009

1. a) To verify the TDMA solver, the previous assignment question was used given again here:

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \\ \phi_5 \end{bmatrix} = \begin{bmatrix} -204 \\ -12 \\ -20 \\ -28 \\ -1236 \end{bmatrix}$$

Using Gauss-Seidel in previous problem, the values of ϕ_i were

$$\phi = \{167, 297, 415, 513, 583\}$$

And using the tdma code developed, the answer was again

$$\phi = \{167, 297, 415, 513, 583\}$$

which is same as the last assignment answer. The code can be run and checked for these output values.

- b) The equations for line-by-line TDMA solver ~~were~~ ^{are} derived first for Dirichlet boundary conditions, which is just temperature at each of the 4-faces.

The governing diffusion equation is $\frac{d}{dx} \left(\Gamma \frac{d\phi}{dx} \right) + S_\phi = 0$

$$\Rightarrow \int_{cv} \nabla \cdot \left(\hat{i} \Gamma \frac{d\phi}{dx} + \hat{j} \Gamma \frac{d\phi}{dy} \right) dv + \int_{cv} S_\phi dv = 0$$

Using GDT,

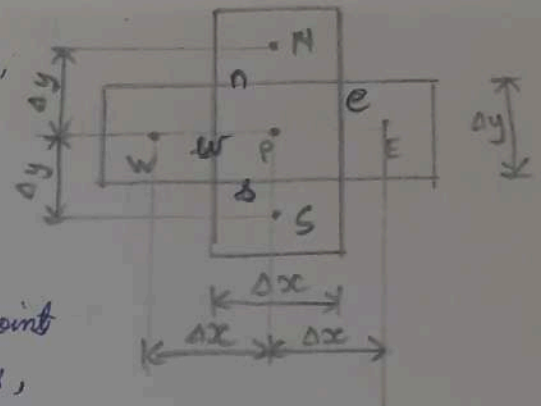
$$\int_{cs} \left(\hat{i} \Gamma \frac{d\phi}{dx} + \hat{j} \Gamma \frac{d\phi}{dy} \right) \cdot d\vec{A} + \int_{cv} S_\phi dv = 0 \quad \text{--- (1)}$$

$$d\vec{A} = \Delta x \hat{i} + \Delta y \hat{j}$$

For a general interior cell as shown in the diagram,

$$Sx_e = Sx_w = \Delta x$$

$$Sy_n = Sy_s = \Delta y$$



Also, since we are going to deal with equal grid points in the 2nd problem along both x and y directions,

$$\Delta x = \Delta y.$$

Also, linearizing the source term gives $S = S_c + S_p \phi_p$

Equation ① ~~becomes~~ simplifies to

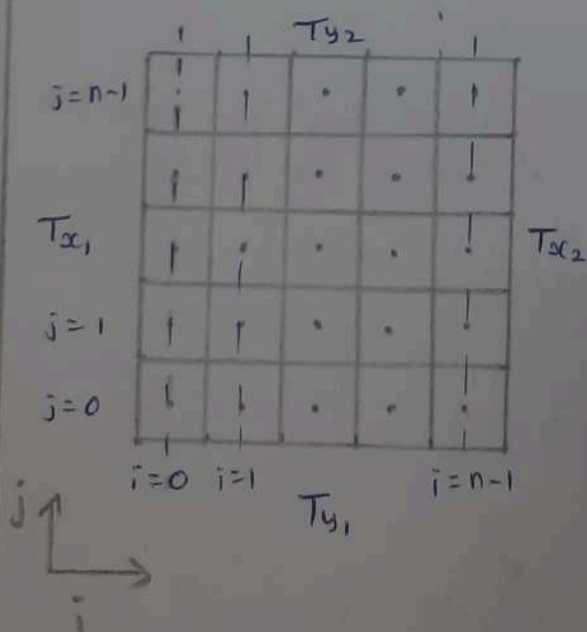
$$\Gamma_e \frac{\partial \phi}{\partial x} \Big|_e \Delta y - \Gamma_w \frac{\partial \phi}{\partial x} \Big|_w \Delta y + \Gamma_n \frac{\partial \phi}{\partial y} \Big|_n \Delta x - \Gamma_s \frac{\partial \phi}{\partial y} \Big|_s \Delta x + (S_c + S_p \phi_p) \Delta V = 0 \quad \text{--- ②}$$

⇒ In the TDMA solver developed, the arguments to the TDMA function are the matrix a , the no. of elements in vector x , and the vector b and the vector x . But the inputs are modified in this program to have only the diagonal elements of a $m \times m$ matrix instead of the entire matrix.

The TDMA solver solves $a\bar{x} = \bar{b}$ and finds $\bar{x}$.

→ $S_p = 0$ assumed.

→ To derive equations specific to cells, the domain is discretized as given below.



→ The bottom left node is represented by $i=0, j=0$ and the remaining nodes: similar to a cartesian coordinate system.

→ The boundary conditions are T_{x1}, T_{x2}, T_{y1} , and T_{y2} as shown in the figure.

For the node $i=0, j=0$,

3

$$\Rightarrow \Gamma_e \frac{\partial \phi}{\partial x}|_e \Delta y - \Gamma_w \frac{\partial \phi}{\partial x}|_w \Delta y + \Gamma_n \frac{\partial \phi}{\partial y}|_n \Delta x - \Gamma_s \frac{\partial \phi}{\partial y}|_s \Delta x + S_c \Delta V = 0 \quad (3)$$

$$\Gamma = \text{constant} = \Gamma_e = \Gamma_w = \Gamma_n = \Gamma_s, \quad \Delta V = (\Delta x)(\Delta y)$$

if

$$\Rightarrow \Gamma \left[\frac{(\phi_E - \phi_P) \Delta y}{\Delta x} - \frac{(\phi_P - \phi_W) \Delta y}{\Delta x} + \frac{(\phi_N - \phi_P) \Delta x}{\Delta y} - \frac{(\phi_P - \phi_S) \Delta x}{\Delta y} \right] + S_c \Delta V = 0 \quad (4)$$

$\Rightarrow \because \Delta x = \Delta y$, above equation becomes

$$\text{In } \phi_E + \phi_W + \phi_N + \phi_S - 4\phi_P = -\frac{S_c (\Delta x)^2}{\Gamma} \quad : \text{For a typical interior cell.}$$

We will be solving TDMA along y-direction, so we will use guess values or previous iteration values of ϕ_E and ϕ_W and solve for ϕ_P .

$$\text{So, } \phi_S - 4\phi_P + \phi_N = -\phi_E - \phi_W - \frac{S_c (\Delta x)^2}{\Gamma}$$

In

we now let $\phi = T$ and $S_c = \frac{S_c (\Delta x)^2}{\Gamma}$ for convenience

$$\Rightarrow T_S - 4T_P + T_N = -(\phi_E + \phi_W) - S_c \quad (5)$$

For all interior cells, the i and j indices will vary from $i=1$ to $n-2$ and $j=1$ to $n-2$. Since these are interior cells, we need not worry about the boundary.

Based on the figure beside, we can write

$$T_P = T_{i,j}, \quad T_N = T_{i,j+1}, \quad T_S = T_{i,j-1},$$

$$T_E = T_{i+1,j}, \quad T_W = T_{i-1,j}$$

Substituting in (5),

$i-1, j+1$	$i, j+1$	$i+1, j+1$
$i-1, j$	i, j	$i+1, j$
$i-1, j-1$	$i, j-1$	$i+1, j-1$

$$T_{i,j-1} - 4T_{i,j} + T_{i,j+1} = - (T_{i+1,j} + T_{i-1,j}) - S_c \quad (6)$$

$$b_{ij} = -(T_{i+1j} + T_{i-1j}) - s_c$$

To complete the matrix for TDMA, we need $i=0$ and $j=n-1$ also because a line contains indices $j=0$ to $j=n-1$.

$y = n - 1$

Diagram illustrating a 2D grid structure with nodes labeled N, W, P, and E. The vertical distance between the bottom row and the top row is indicated as $\frac{\Delta x}{2}$. The label T_{y_i} is shown below the grid.

$$\Rightarrow T_N - 5T_P = -2T_y, -(T_E + T_W) - S_C$$

$$T_{12} + T_{12} = 2T_{12}$$

$$\Rightarrow \boxed{T_{i1} - 5T_{i0} = b_{i0}} \quad - (7)$$

$$T_{in-2} - 5T_{in-1} = b_{in-1} \quad (8)$$
$$\begin{bmatrix} 1 & -5 & 0 & 0 & 0 \\ 0 & 1 & -4 & 1 & 0 \\ 0 & 0 & 1 & -4 & 1 \\ 0 & 0 & 0 & 1 & -5 \end{bmatrix} \begin{bmatrix} T_{i0} \\ T_{i1} \\ \vdots \\ T_{in-1} \end{bmatrix} = \begin{bmatrix} b_{i0} \\ b_{i1} \\ \vdots \\ b_{in-1} \end{bmatrix} \quad - (9)$$

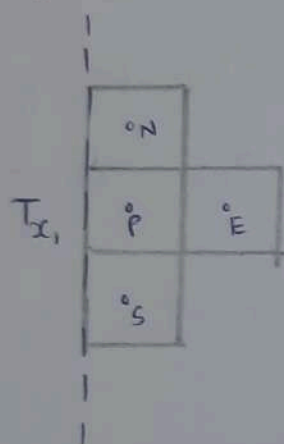
where $b_{i0} = -\cancel{5T_{i0}} + \cancel{T_{i1}} - 2T_{y1} - (T_{i+10} + T_{i-10}) - S_c$

$b_{i,n-1} = \cancel{T_{i,n-2}} - \cancel{5T_{i,n-1}} - 2T_{y2} - (T_{i+1,n-1} + T_{i-1,n-1}) - S_c$

$b_{ij} = -\cancel{(T_{i+1j})} - (T_{i+1j} + T_{i-1j}) - S_c$

which is valid for $i=1$ to $n-2$.

Now for $i=0$, $j=1$ to $n-1$ represents the left most nodes except the top left and bottom left corner nodes.



Eqⁿ (2) $\Rightarrow T_E - T_P + T_N - T_P + T_S - T_P + \frac{T_{x1} - T_P}{1/2} = -S_c$

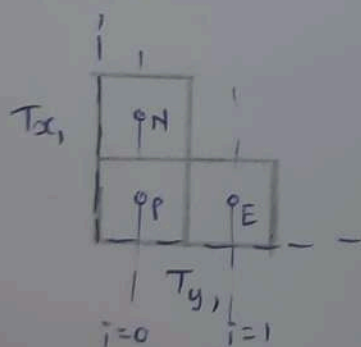
$\Rightarrow 4T_S - 5T_P + T_N = -2T_{x1} - T_E - S_c$

In index notation,

$T_{0,j-1} - 5T_{0,j} + T_{0,j+1} = \underbrace{-2T_{x1} - T_{1,j}}_{b_{1j}} - S_c \quad (10)$

As before, to complete the matrix, we need $j=0$ and $j=n-1$ values also.

$i=0, j=0$



Eqⁿ (2) $\Rightarrow T_E + T_N$

$T_E - T_P + T_N - T_P + \frac{T_{y1} - T_P}{1/2} + \frac{T_{x1} - T_P}{1/2} = -S_c$

$\Rightarrow -6T_P + T_N = -T_E - (T_{y1} + T_{x1}) - S_c$

$\Rightarrow -6T_{00} + T_{01} = \underbrace{-T_{10} - (T_{x1} + T_{y1})}_{b_{00}} - S_c \quad (11)$

Similarly, for top-left corner,

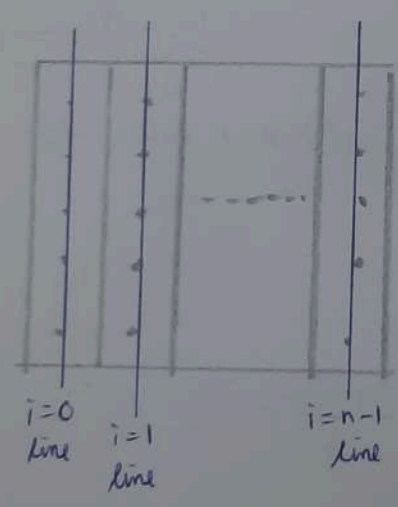
$T_{0,n-2} - 6T_{0,n-1} + T_{0n} = \underbrace{-T_{1,n-1} - (T_{x1} + T_{y2})}_{b_{0,n-1}} - S_c \quad (12)$

From eqns (10), (11) and (12), we get for left most line-nodes

$$\begin{bmatrix} -6 & 1 & 0 & 0 & 0 \\ 1 & -5 & 1 & 0 & 0 \\ 0 & 1 & -5 & 1 & 0 \\ 0 & 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} T_{00} \\ T_{01} \\ \vdots \\ T_{0,n-1} \end{bmatrix} = \begin{bmatrix} b_{00} \\ b_{01} \\ \vdots \\ b_{0,n-1} \end{bmatrix} \quad \text{where} \quad \begin{aligned} b_{00} &= -T_{10} - (T_{x1} + T_{y1}) - S_c \\ b_{0,n-1} &= -T_{1,n-1} - (T_{x1} + T_{y2}) - S_c \\ b_{0j} &= -2T_{x1} - 4T_{1j} - S_c \end{aligned} \quad (13)$$

and similarly for the right most column ($i=n-1$), we get

$$\begin{bmatrix} -6 & 1 & 0 & 0 & 0 \\ 1 & -5 & 1 & 0 & 0 \\ 0 & 1 & -5 & 1 & 0 \\ 0 & 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} T_{n-1,0} \\ T_{n-1,1} \\ \vdots \\ T_{n-1,n-1} \end{bmatrix} = \begin{bmatrix} b_{n-1,0} \\ b_{n-1,1} \\ \vdots \\ b_{n-1,n-1} \end{bmatrix} \quad \text{where} \quad \begin{aligned} b_{n-1,0} &= -T_{n-2,0} - (T_{x2} + T_{y1}) - S_c \\ b_{n-1,n-1} &= -T_{n-2,n-1} - (T_{x2} + T_{y2}) - S_c \\ b_{n-1j} &= -T_{x2} - T_{n-2j} - S_c \end{aligned} \quad (14)$$



From (9), (13) and (14),
 $i=0$ and $i=n-1$ lines have diagonal elements
 as $\{-6, -5, -5, \dots, -6\} = d[0]$
 while other i have diagonal elements
 as $\{-5, -4, -4, \dots, -5\} = d[1]$

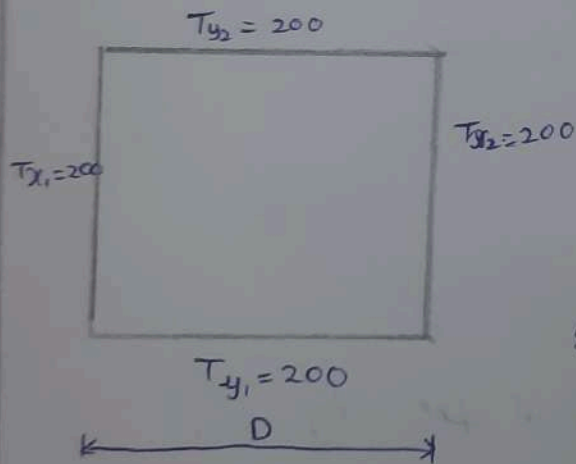
- The ~~tdma~~ TDMA function ~~tdma~~ is called as $tdma(n, d[0], T[0:n-1], b[0:n-1])$ for boundary elements which solves for $T[0][j]$, ($i=0$ ~~column~~ line) and $T[n-1][j]$, ($i=n-1$ line). For interior ~~elements~~ ^{lines}, it is called as $tdma(n, d[1], T[i], b[i])$.
- only diagonal elements are passed because the other elements are 1 in each tridiagonal matrix.

* Solution procedure

- 1) First guess the value of T
- 2) Initialize matrix b and diagonal matrix d with guess T and $B(3)$.
- 3) ~~to solve~~ ^{use} TDMA to solve for $T[i]$
- 4) Update b to include $T[i]$ values
- 5) use TDMA to solve for $T[i+1]$
- 6) Repeat 3), 4) and 5) for $i=0$ to $n-2$.
- 7) First iteration done
- 8) Iterate until $T - T^* < \epsilon$, $\epsilon = \text{residue} (=10^{-3} \text{ in code})$
- 9) once converged, plot $T[i][j]$

→ Verification ~~first check~~ problems ^{are} given below

a)



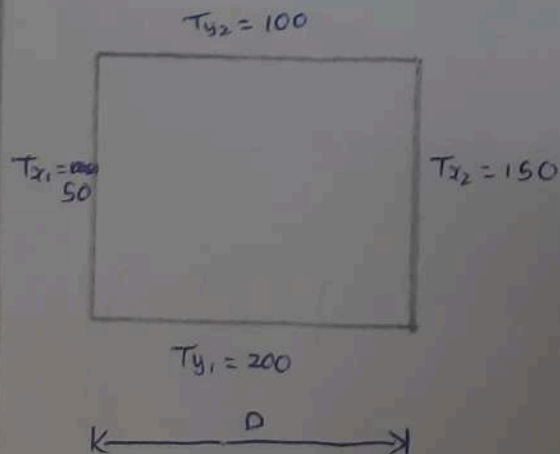
guess value = 190

$D = 1.0$, $T = 0.01$, $S = 0$

$n = 20$, no. of iterations = 1000.

Solution converged after ²⁴¹ ~~200~~ iterations and same temperature everywhere found to be 200.

b)



guess value = 190

~~n=20~~

Solution converged after 278 iterations.

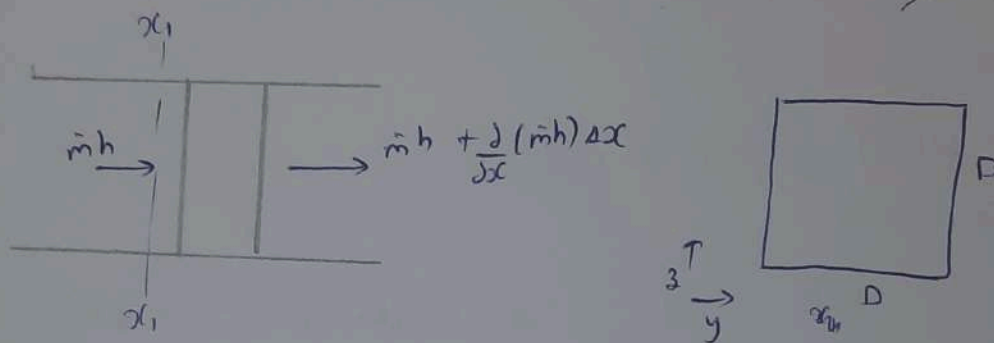
Fesh

2.

2. Non-Newtonian Square Duct Floor

a) We use a similar procedure for discretization, but before that,

$$\frac{\partial}{\partial x} (844 \rho T) = \nabla \cdot (k \nabla T) = \frac{\partial}{\partial y} \left(\frac{k \partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{k \partial T}{\partial z} \right) + \frac{\partial}{\partial x} \left(\frac{k \partial T}{\partial x} \right)$$

Energy balance yields $\dot{m}h + q'' p \Delta x - \dot{m}h - \frac{\partial}{\partial x} (\dot{m}h) \Delta x = 0$

$$\Rightarrow \frac{d}{dx} (844 D^2 \rho T) = q'' / (4D)$$

$$\Rightarrow \frac{dT}{dx} = \frac{4q''}{844 D} \text{ which is constant.}$$

⇒ the governing equation is

$$\frac{\partial}{\partial x} (844 \rho \left(\frac{4q''}{844 D} \right)) = \nabla \cdot (k \nabla T)$$

$$\Rightarrow \nabla \cdot (k \nabla T) - \frac{4q''}{D} = 0 \rightarrow \text{Integrate and apply GDT}$$

after GDT and,

$$\text{Source term is } -\frac{4q'' \Delta V}{D} \Rightarrow S_c = -\frac{4q'' \Delta V}{D \Gamma} \text{ and } q_m'' = \frac{q'' dx}{\Gamma}$$

b) The only change that comes in this problem is the Neumann BCs. So, only the matrices and the ~~to~~ change as given below.

$$\text{For } i=0 \text{ line, } \begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 1 & -3 & 1 & 0 & 0 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} T_{01} \\ T_{02} \\ \vdots \\ T_{0n-1} \end{bmatrix} = \begin{bmatrix} 2q'' - T_{10} - S_c \\ q'' - T_{11} - S_c \\ \vdots \\ 2q'' - T_{1n-1} - S_c \end{bmatrix} = \begin{bmatrix} b_{01} \\ b_{02} \\ \vdots \\ b_{0n-1} \end{bmatrix}$$

For ~~each~~ line
 $i = n-1$

$$\begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 1 & -3 & 1 & 0 & 0 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} T_{n-10} \\ T_{n-11} \\ \vdots \\ T_{n-1n-1} \end{bmatrix} = \begin{bmatrix} 2q - T_{n-20} - S_c \\ q - T_{n-21} - S_c \\ \vdots \\ 2q - T_{n-2n-2} - S_c \end{bmatrix} = \begin{bmatrix} b_{n-10} \\ b_{n-11} \\ \vdots \\ b_{n-1n-1} \end{bmatrix}$$

And for $i = 1$ to $n-2$ lines,

$$\begin{bmatrix} -3 & 1 & 0 & 0 & 0 \\ 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & -4 & 1 & 0 \\ 0 & 0 & 1 & -4 & 1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} T_{i0} \\ T_{i1} \\ \vdots \\ T_{in-1} \end{bmatrix} = \begin{bmatrix} q - (T_{i+10} + T_{i-10}) - S_c \\ - (T_{i+11} + T_{i-11}) - S_c \\ \vdots \\ q - (T_{i+1n-1} + T_{i-1n-1}) - S_c \end{bmatrix} = \begin{bmatrix} b_{i0} \\ b_{i1} \\ \vdots \\ b_{in-1} \end{bmatrix}$$

The solution procedure is same as the previous question.

c) $T_b = \frac{\iint u T \, dy \, dz}{\iint u \, dy \, dz} = \frac{\iint T \, dy \, dz}{\iint dy \, dz}$ since $u = \text{constant}$.

d)
$$Nu = \frac{q_- D/k}{(\bar{T}_w - T_b)}$$

For $q_- = 100 \, \text{W/m}^2$,

$n=10 \Rightarrow Nu = 8.333$

$n=20 \Rightarrow Nu = 7.020$

$n=30 \Rightarrow Nu = 6.747$

e) Inversion is $O((n+1)!)$ operation. To obtain solution for any smaller sized matrix, it can be done but it will take long time for matrices of size more than 10×10 . ~~So~~ we would be able to obtain a solution but at very high computational cost which can be ~~avoided~~ avoided by using LBL-TDMA ~~in~~ which requires lesser computational cost.

f) with different initial guesses, the temperature distribution remains the same, but the actual temperature scale varies. The temperatures kind of scale with the initial guess which can be avoided by fixing one temperature node. Also, the number of iterations required changes slightly with the different guess values of 100, 200 and 300K.

Initial guess value	Temperature scale
100 K	98 to 103 K
200 K	198 to 203 K
300 K	298 to 303 K

For
 $\leftarrow n=30$